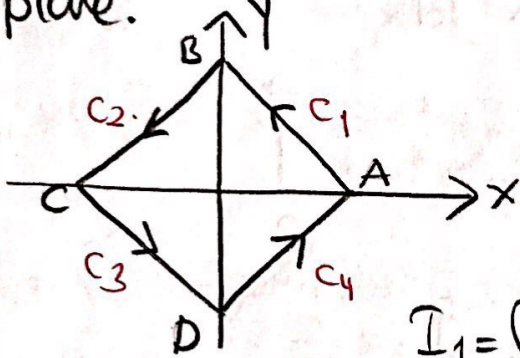


Exercises

1) Evaluate $I = \oint_C [(x-y^3)dx + x^2ydy]$ where C is the closed curve: $|x|+|y|=1$ on xy -plane.



$$\textcircled{*} C_1 : \overline{AB} : \begin{cases} x+y=1 \\ x=x \\ y=1-x \end{cases} \begin{cases} dx=dx \\ dy=-dx \end{cases}$$

$$I_1 = \int_{C_1} (x-y^3)dx + x^2ydy$$

$$I_1 = \int_1^0 [x - (1-x)^3]dx + x^2(1-x)(-dx)$$

$$I_1 = \int_1^0 [x - (1 - 3x + 3x^2 - x^3) - x^2(1-x)]dx$$

$$I_1 = \int_1^0 (x - 1 + 3x - 3x^2 + \underline{x^3} - x^2 + \underline{x^3})dx$$

$$I_1 = \int_1^0 (2x^3 - 4x^2 + 4x - 1)dx$$

$$I_1 = \left. \frac{1}{2}x^4 - \frac{4}{3}x^3 + 2x^2 - x \right|_1^0 = 0 - \left(\frac{1}{2} - \frac{4}{3} + 2 - 1 \right) = -\frac{1}{6}$$

$$\underline{I_1 = -1/6}$$

$$\textcircled{*} C_2 : \overline{BC} : \begin{cases} -x+y=1 \\ x=x \\ y=1+x \end{cases} \begin{cases} dx=dx \\ dy=dx \end{cases}$$

$$I_2 = \int_{C_2} (x - y^3) dx + x^2 y dy = \int_0^{-1} [x - (x+1)^3] dx + x^2(x+1) dx$$

$$I_2 = \int_0^{-1} (x - (x^3 + 3x^2 + 3x + 1) + x^3 + x^2) dx$$

$$I_2 = \int_0^{-1} (x - \cancel{x^3} - 3x^2 - 3x - 1 + \cancel{x^3} + x^2) dx$$

$$I_2 = \int_0^{-1} (-2x^2 - 2x - 1) dx = -\frac{2}{3}x^3 - x^2 - x \Big|_0^{-1}$$

$$I_2 = -\frac{2}{3}(-1) - 1 + 1 = \frac{2}{3} \Rightarrow \underline{I_2 = \frac{2}{3}}$$

$$\otimes C_3: \overline{CD}: -x - y = 1$$

$$\left. \begin{array}{l} x = x \\ y = -x - 1 \end{array} \right\} \begin{array}{l} dx = dx \\ dy = -dx \end{array}$$

$$I_3 = \int_{C_3} (x - y^3) dx + x^2 y dy = \int_{-1}^0 [x - (-x-1)^3] dx + x^2(-x-1)(-dx)$$

$$I_3 = \int_{-1}^0 [x + (x^3 + 3x^2 + 3x + 1) + x^2(x+1)] dx$$

$$I_3 = \int_{-1}^0 [x + \underline{x^3} + 3x^2 + 3x + 1 + \underline{x^3} + x^2] dx$$

$$I_3 = \int_{-1}^0 (2x^3 + 4x^2 + 4x + 1) dx = \frac{x^4}{2} + \frac{4}{3}x^3 + 2x^2 + x \Big|_{-1}^0$$

$$I_3 = 0 - \left(\frac{1}{2} - \frac{4}{3} + 2 - 1 \right) = -\frac{1}{6}$$

$$\underline{I_3 = -\frac{1}{6}}$$

$$\textcircled{\times} C_4: \overline{DA}: x-y=1$$

$$\left. \begin{array}{l} x=x \\ y=x-1 \end{array} \right\} \begin{array}{l} dx=dx \\ dy=dx \end{array}$$

$$I_4 = \int_{C_4} (x-y^3)dx + x^2ydy = \int_0^1 (x - (x-1)^3)dx + x^2(x-1)dx$$

$$I_4 = \int_0^1 [x - (x^3 - 3x^2 + 3x - 1) + x^3 - x^2]dx$$

$$I_4 = \int_0^1 (x - \cancel{x^3} + 3x^2 - 3x + 1 + \cancel{x^3} - x^2)dx$$

$$I_4 = \int_0^1 (2x^2 - 2x + 1)dx = \left. \frac{2}{3}x^3 - x^2 + x \right|_0^1 = \frac{2}{3} - 1 + 1$$

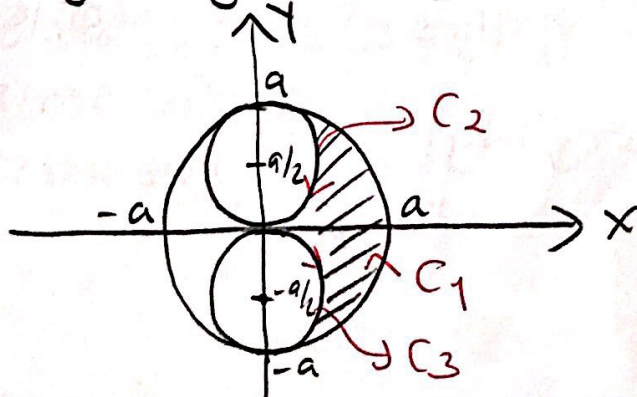
$$\underline{I_4 = \frac{2}{3}}$$

$$I = I_1 + I_2 + I_3 + I_4 = -\frac{1}{6} + \frac{2}{3} - \frac{1}{6} + \frac{2}{3} = 1 //$$

2) Evaluate the integral $I = \oint_C (y^2 - xy)dx + (x^2 - xy)dy$ on C which is enclosing the closed region that is between the circles $x^2 + y^2 = a^2$, $x^2 + y^2 + ay = 0$, $x \geq 0$.

$$x^2 + y^2 + ay = 0 \Rightarrow x^2 + (y + \frac{a}{2}) = \frac{a^2}{4}$$

$$x^2 + y^2 - ay = 0 \Rightarrow x^2 + (y - \frac{a}{2}) = \frac{a^2}{4}$$



$$C_1: \begin{cases} x^2 + y^2 = a^2 \\ x = a \cos \theta \\ y = a \sin \theta \end{cases} \Rightarrow \begin{cases} dx = -a \sin \theta d\theta \\ dy = a \cos \theta d\theta \end{cases}$$

$$I_1 = \int_{C_1} (y^2 - xy) dx + (x^2 - xy) dy$$

$$I_1 = \int_{-\pi/2}^{\pi/2} (a^2 \sin^2 \theta - a^2 \sin \theta \cos \theta)(-a \sin \theta d\theta) + \int_{-\pi/2}^{\pi/2} (a^2 \cos^2 \theta - a^2 \sin \theta \cos \theta)(a \cos \theta d\theta)$$

$$I_1 = \int_{-\pi/2}^{\pi/2} (-a^3 \sin^3 \theta + a^3 \sin^2 \theta \cos \theta + a^3 \cos^3 \theta - a^3 \sin \theta \cos^2 \theta) d\theta$$

$$I_1 = \int_{-\pi/2}^{\pi/2} [-a^3 (1 - \cos^2 \theta) \sin \theta + a^3 \sin^2 \theta \cos \theta + a^3 \cos \theta (1 - \sin^2 \theta) - a^3 \sin \theta \cos^2 \theta] d\theta$$

$$I_1 = \int_{-\pi/2}^{\pi/2} [-a^3 \sin \theta + a^3 \cos^2 \theta \sin \theta + a^3 \sin^2 \theta \cos \theta + a^3 \cos \theta - a^3 \cos \theta \sin^2 \theta - a^3 \sin \theta \cos^2 \theta] d\theta$$

$$I_1 = \int_{-\pi/2}^{\pi/2} a^3 (\cos \theta - \sin \theta) d\theta = a^3 [\sin \theta + \cos \theta] \Big|_{-\pi/2}^{\pi/2}$$

$$I_1 = a^3 [(1+0) - (-1+0)] = 2a^3 //$$

$$C_2: x^2 + y^2 = ay \Rightarrow r^2 = a r \sin \theta \Rightarrow r = a \sin \theta$$

$$x = r \cos \theta = a \sin \theta \cos \theta \Rightarrow dx = (a \cos^2 \theta - a \sin^2 \theta) d\theta$$

$$y = r \sin \theta = a \sin^2 \theta \Rightarrow dy = a \cdot 2 \sin \theta \cos \theta d\theta$$

$$I_2 = \int_{C_2} (y^2 - xy) dx + (x^2 - xy) dy$$

$$(*) \begin{cases} dx = a \cos 2\theta d\theta \\ dy = a \sin 2\theta d\theta \end{cases}$$

$$I_2 = \int_{\pi/2}^0 (a^2 \sin^4 \theta - a^2 \sin^2 \theta \cos \theta) a (\cos^2 \theta - \sin^2 \theta) d\theta + (a^2 \sin^2 \theta \cos^2 \theta - a^2 \sin^3 \theta \cos \theta) 2a \sin \theta \cos \theta d\theta$$

$$I_2 = \int_{\pi/2}^0 [a^3 (\sin^4 \theta \cos^2 \theta - \sin^4 \theta \sin^2 \theta - \sin^3 \theta \cos^3 \theta + \sin^4 \theta \cos \theta) + 2a^3 (\sin^2 \theta \cos^2 \theta - \sin^3 \theta \cos \theta) \sin \theta \cos \theta] d\theta$$

$$\vdots$$

$$I_2 = \int_{\pi/2}^0 a^3 (\cos \theta \sin^3 \theta - \sin^4 \theta) d\theta = a^3 \left(\frac{3\pi+4}{16} \right)$$

$$C_3: x^2 + y^2 = -ay \Rightarrow r^2 = -ar \sin \theta \Rightarrow r = -a \sin \theta$$

$$\begin{cases} x = r \cos \theta = -a \sin \theta \cos \theta \\ y = r \sin \theta = -a \sin^2 \theta \end{cases} \begin{cases} dx = -a \cos 2\theta d\theta \\ dy = -a \sin 2\theta d\theta \end{cases}$$

$$I_3 = \int_{C_3} (y^2 - xy) dx + (x^2 - xy) dy$$

$$I_3 = \int_0^{-\pi/2} (a^2 \sin^4 \theta - a^2 \sin^3 \theta \cos \theta) (-a \cos 2\theta) d\theta + (a^2 \sin^2 \theta \cos^2 \theta - a^2 \sin^3 \theta \cos \theta) (-a \sin 2\theta) d\theta$$

$$I_3 = \int_0^{-\pi/2} a^3 (\sin^4 \theta - \sin^3 \theta \cos \theta) d\theta = -a^3 \left(\frac{3\pi+4}{16} \right)$$

$$I = I_1 + I_2 + I_3$$

$$I = 2a^3 + a^3 \left(\frac{3\pi+4}{16} \right) - a^3 \left(\frac{3\pi+4}{16} \right) = 2a^3$$

3) $\vec{V} = (x^2 + 5\lambda y + 3yz)\vec{i} + (5x + 3\lambda xz - 2)\vec{j} + [(2+\lambda)xy - 4z]\vec{k}$ is the power applied from the point $A(0,0,0)$ to $B(1,2,3)$. For which value of λ this work is path-independent? Find the value of λ and then evaluate the work by using potential function.

If $\text{Rot } \vec{V} = 0$ then it's path-independent.

$$\text{Rot } \vec{V} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 + 5\lambda y + 3yz & 5x + 3\lambda xz - 2 & (2+\lambda)xy - 4z \end{vmatrix} = 0$$

$$\text{Rot } \vec{V} = \underbrace{[(2+\lambda)x - 3\lambda x]}_0 \vec{i} + \underbrace{[3y - (2+\lambda)y]}_0 \vec{j} + \underbrace{[5 + 3\lambda z - (5\lambda + 3z)]}_0 \vec{k}$$

$$\left. \begin{array}{l} 2 + \lambda - 3\lambda = 0 \\ 3 - 2 - \lambda = 0 \\ 1 - \lambda = 0 \end{array} \right\} \boxed{\lambda = 1}$$

For $\lambda = 1$ $\text{Rot } \vec{V} = 0 \Rightarrow I = \int_{(A)}^{(B)} \vec{V} d\vec{r}$ is path-ind.

$$I = \int_{(A)}^{(B)} \underbrace{(x^2 + 5y + 3yz)}_P dx + \underbrace{(5x + 3xz - 2)}_Q dy + \underbrace{(3xy - 4z)}_R dz$$

Then there exist a function $u = u(x, y, z)$ such that;

$$\frac{\partial u}{\partial x} = x^2 + 5y + 3yz, \quad \frac{\partial u}{\partial y} = 5x + 3xz - 2, \quad \frac{\partial u}{\partial z} = 3xy - 4z$$

$$u(x, y, z) = \frac{x^3}{3} + 5xy + 3yzx + h(y, z)$$

$$\frac{\partial u}{\partial y} = \cancel{5x} + \cancel{3zx} + \frac{\partial h}{\partial y} = \cancel{5x} + \cancel{3xz} - 2$$

$$\frac{\partial u}{\partial y} = -2 \Rightarrow h(y, z) = -2y + \alpha(z)$$

$$u(x, y, z) = \frac{x^3}{3} + 5xy + 3yz - 2y + \alpha(z)$$

$$\frac{\partial u}{\partial z} = \cancel{3yx} + \frac{d\alpha}{dz} = \cancel{3yx} - 4z \Rightarrow \frac{d\alpha}{dz} = -4z$$

$$\alpha(z) = -2z^2 + C$$

$$u(x, y, z) = \frac{x^3}{3} + 5xy + 3xyz - 2y - 2z^2 + C$$

$$I = \int_{(A)}^{(B)} du = u(x, y, z) \Big|_{(A)}^{(B)}$$

$$I = \frac{x^3}{3} + 5xy + 3xyz - 2y - 2z^2 + C \Big|_{A(0,0,0)}^{B(1,2,3)}$$

$$I = \frac{1}{3} + 10 + \cancel{18} - 4 - \cancel{18} + \cancel{C} - \cancel{C}$$

$$\underline{I = 19/3}$$